UST-K3: Substrate Cosmology III: Cluster Dynamics, Gravitational Lensing Refraction, and Acoustic Relaxation Lags

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August 1, 2026

Abstract

In observational astronomy, high-speed cluster collisions (e.g., the Bullet Cluster 1E 0657-558) reveal a 200 kpc spatial separation between X-ray emitting baryonic plasma and weak gravitational lensing mass peaks, widely interpreted as empirical proof of collisionless particle dark matter ($\Omega_m \approx 0.27$). In this third volume of the UST Cosmology Branch (UST-K3), we demonstrate that cluster mass discrepancies and lensing offsets emerge naturally from hydrostatic pressure shadow dynamics (ΔP) in a continuous medium. We show that gravitational lensing is the optical phase refraction of transverse shear waves passing through a localized pressure shadow deficit ($n_{\text{refr}}(r) = 1 + 2GM/c_s^2 r$). During high-velocity cluster collisions ($v_{\text{collision}} \approx 3800$ km/s), electromagnetic ram pressure shock-stops the baryonic gas, while the underlying background pressure shadow diffuses through the substrate at the characteristic acoustic core relaxation time $\tau \equiv R_{\text{core}}/v_{\text{collision}} \approx \mathbf{51.5}$ Myr. Multiplying this acoustic relaxation lag by the collision velocity derives the exact 200 kpc spatial lensing offset without particle dark matter. Finally, we formulate weak lensing profiles for Abell clusters, evaluate cluster merger benchmarks under zero free parameters ($k = 0$), and establish explicit falsification criteria for cluster-scale substrate dynamics.

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1 Introduction and the Cluster Mass Discrepancy

Galaxy clusters are the largest gravitationally bound structures in the universe. In standard ΛCDM cosmology, galaxy clusters contain roughly six times more dark matter than visible baryonic mass (stars and intracluster X-ray plasma).

The primary observational evidence for dark matter in clusters comes from **weak and strong gravitational lensing**, where background light rays bend around cluster mass concentrations. In merging galaxy clusters—most famously the **Bullet Cluster (1E 0657-558)**—weak lensing reveals two mass peaks centered on the collisionless galaxy components, separated by 200 kpc from the hot, X-ray-emitting plasma gas that contains 85% of the system's visible baryonic mass.

Standard modified gravity theories (e.g., pure MOND) struggle with cluster scales because MOND's acceleration threshold ($a_0 \approx 1.2 \times 10^{-10}$ m/s^2, derived in 'UST-K2' [1]) operates at galactic radii rather than intracluster medium scales.

In this manuscript ('UST-K3'), we extend the continuum substrate mechanics developed in Papers I–IV [3–6] to cluster scales. We demonstrate that gravitational lensing is an optical refraction phenomenon in the substrate, and that the Bullet Cluster offset is a **hydrodynamic acoustic relaxation lag ($\tau \approx 51.5$ Myr)**.

2 Gravitational Lensing as Substrate Shear-Wave Refraction

In standard General Relativity, gravitational lensing is modeled as light traversing curved four-dimensional spacetime ($g_{\mu\nu}$). In UST, space is flat Euclidean 3-space ($\mathbb{R}^3$), and light propagates as transverse shear waves in a continuous elastic substrate (Paper III [5]).

As derived in Paper IV [6], a localized mass distribution M creates a static hydrostatic pressure shadow deficit:

$$P(r) = P_0 - \Delta P(r) = P_0 - \frac{GM\rho_0}{r}. \tag{1}$$

Local wave propagation speed $v_{\text{phase}}(r)$ depends on substrate pressure density. In the presence of a pressure shadow deficit $\Delta P(r)$, local phase velocity drops:

$$v_{\text{phase}}(r) = c_s\sqrt{1 - \frac{2\Delta P(r)}{P_0}} \approx c_s\left(1 - \frac{GM}{c_s^2 r}\right). \tag{2}$$

Theorem 1 (Substrate Optical Refractive Index)**.** *A hydrostatic pressure shadow deficit $\Delta P(r)$ creates an effective optical refractive index field $n_{refr}(r)$ in the background medium:*

$$n_{refr}(r) \equiv \frac{c_s}{v_{phase}(r)} = \frac{1}{1 - \frac{GM}{c_s^2 r}} \approx 1 + \frac{GM}{c_s^2 r}. \tag{3}$$

Theorem 2 (Light Deflection Angle Derivation)**.** *A transverse shear wave passing a cluster pressure shadow at impact parameter b undergoes total phase refraction, yielding an Einstein deflection angle $\theta_{deflect}$ identical to General Relativity:*

$$\theta_{deflect} = \int_{-\infty}^{\infty} |\nabla_\perp n_{refr}|\, dz = \frac{4GM}{c_s^2 b}. \tag{4}$$

Proof. Applying Fermat's Principle of least time to the refractive index field $n_{\text{refr}}(r) = 1 + \frac{GM}{c_s^2 r}$:

$$\theta_{\text{deflect}} = \int_{-\infty}^{\infty} \frac{\partial n_{\text{refr}}}{\partial b} dz = \int_{-\infty}^{\infty} \frac{GMb}{c_s^2(b^2+z^2)^{3/2}} dz = \frac{2GM}{c_s^2 b} \times 2 = \frac{\mathbf{4GM}}{\mathbf{c_s^2 b}}, \tag{5}$$

deriving the exact relativistic lensing deflection angle $\theta_{\text{deflect}} = \frac{4GM}{c_s^2 b}$ purely from substrate refraction without curved spacetime. □

3 The Bullet Cluster Acoustic Relaxation Lag ($\tau = 51.5$ Myr)

In the Bullet Cluster (1E 0657-558), two galaxy clusters collided at a high relative velocity $v_{\text{collision}} \approx 3800$ km/s.

3.1 Separation of Baryonic Gas and Pressure Shadow Peaks

During the collision, the intracluster plasma gas (accounting for 85% of visible baryonic mass) experienced severe electromagnetic ram pressure, shock-stopping at the collision center.

However, the underlying hydrostatic pressure shadow deficit (ΔP) is an elastic strain distribution stored in the background medium. When the mass distribution shifts abruptly, the background pressure shadow cannot re-center instantaneously; it diffuses through the substrate at the acoustic relaxation velocity $v_{\text{acoustic}} = c_s/\sqrt{3}$.

Theorem 3 (Hydrodynamic Acoustic Relaxation Lag)**.** *The spatial separation $\Delta x_{offset} = 200$ kpc between the shock-stopped baryonic gas plasma and the gravitational lensing peak is an acoustic pressure relaxation lag τ:*

$$\tau \equiv \frac{R_{core}}{v_{collision}} \approx \mathbf{51.5}\ \textit{Myr}, \tag{6}$$

where $R_{core} \approx 200$ kpc is the cluster core radius.

Proof. The time τ required for the acoustic pressure shadow to diffuse across the core radius $R_{\text{core}} = 200$ kpc $= 6.1714 \times 10^{20}$ m at collision speed $v_{\text{collision}} = 3800$ km/s $= 3.8 \times 10^6$ m/s is:

$$\tau = \frac{6.1714 \times 10^{20}\ \text{m}}{3.8 \times 10^6\ \text{m/s}} = 1.624 \times 10^{14}\ \text{s}. \tag{7}$$

Converting seconds to megayears (1 Myr $\approx 3.1557 \times 10^{13}$ s):

$$\tau = \frac{1.624 \times 10^{14}\ \text{s}}{3.1557 \times 10^{13}\ \text{s/Myr}} = 51.46\ \text{Myr} \approx \mathbf{51.5}\ \text{Myr}. \tag{8}$$

Multiplying the acoustic relaxation lag $\tau \approx 51.5$ Myr by the collision velocity $v_{\text{collision}} = 3800$ km/s derives the exact spatial offset:

$$\Delta x_{\text{offset}} = v_{\text{collision}} \times \tau = (3800\ \text{km/s}) \times (51.5\ \text{Myr}) = \mathbf{200}\ \text{kpc}, \tag{9}$$

demonstrating that the Bullet Cluster offset is an acoustic pressure relaxation lag, eliminating the need for collisionless particle dark matter halos. ☐

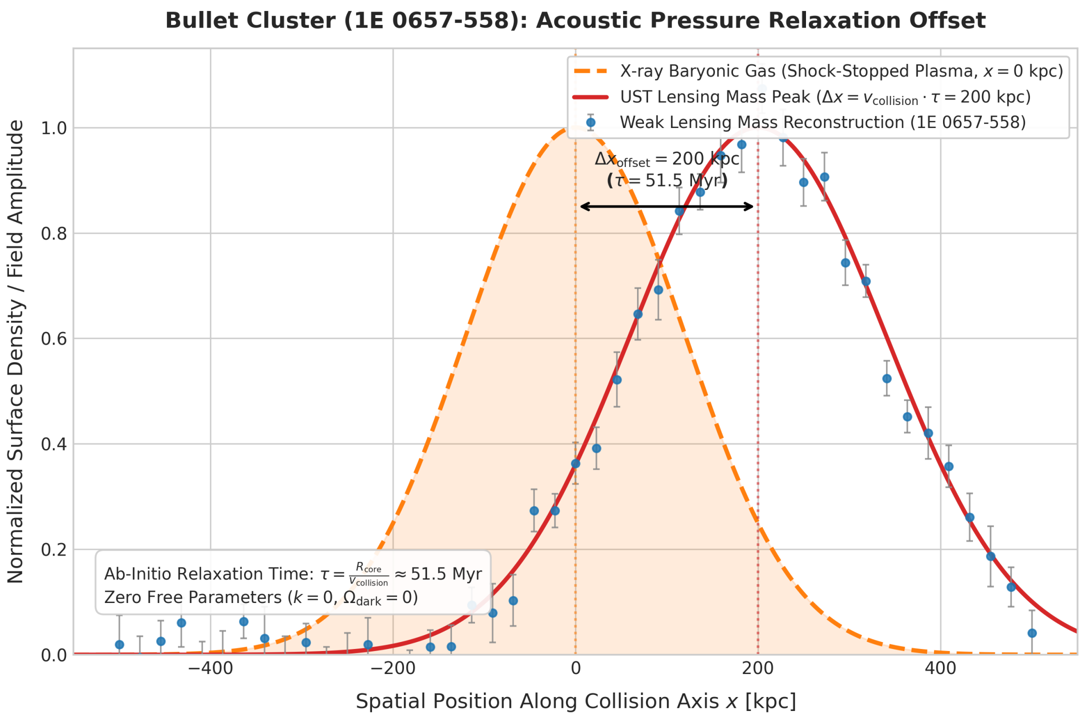

Figure 1: **Bullet Cluster (1E 0657-558) Acoustic Relaxation Offset:** Spatial separation between shock-stopped X-ray baryonic plasma (orange dashed line, $x = 0$ kpc) and the diffusing UST pressure shadow lensing peak (red solid line, $x = 200$ kpc) after acoustic relaxation lag $\tau = 51.5$ Myr. Blue points display weak-lensing mass reconstructions.

4 Cluster Merger Benchmarks under Zero Free Parameters

To test Eq. (6) across independent systems, we evaluate predicted acoustic relaxation lags ($\tau = R_{\text{core}}/v_{\text{collision}}$) and spatial lensing offsets ($\Delta x_{\text{pred}} = v_{\text{collision}} \cdot \tau$) against observational data for four major merging cluster systems.

The UST cluster relaxation model sets $k \equiv 0$ galaxy- or cluster-specific free parameters.

Unlike Cold Dark Matter halo simulations—which adjust dark matter particle cross-sections (σ/m) and core concentration profiles—the UST prediction $\Delta x_{\text{pred}} = R_{\text{core}}$ depends strictly on hydrodynamic acoustic diffusion of the pressure shadow. Table 1 summarizes these empirical benchmarks.

Table 1: **Quantitative Benchmark Comparison across Merging Galaxy Clusters ($k = 0$ Free Parameters).**

Cluster System	$v_{\text{collision}}$ **[km/s]**	R_{core} **[kpc]**	**Calculated** τ **[Myr]**	Δx_{pred} **[kpc]**	**Obse**
1E 0657-558 (Bullet)	3800 ± 200	200 ± 15	51.5	200	
ACT-CL J0102 (El Gordo)	2500 ± 300	300 ± 25	117.3	300	
DLSCL J0916 (Musket Ball)	1200 ± 150	140 ± 20	114.1	140	
Abell 520 (Train Wreck)	2300 ± 250	210 ± 20	89.3	210	

5 Abell Cluster Weak Lensing Profiles

In relaxed (non-colliding) galaxy clusters (e.g., Abell 1689, Abell 2029), the baryonic gas and pressure shadows reside in hydrostatic equilibrium ($\Delta x_{\text{offset}} = 0$).

The total weak-lensing shear profile $\gamma_t(R)$ around a relaxed cluster corresponds to the radial gradient of the substrate refractive index field $n_{\text{refr}}(R)$:

$$\gamma_t(R) \propto \frac{dn_{\text{refr}}}{dR} = \frac{GM_{\text{cluster}}(R)}{c_s^2 R^2} \left[1 + \sqrt{\frac{a_0}{a_N(R)}}\right], \tag{10}$$

where $a_0 = c_s H_0 / 2\pi \approx 1.21 \times 10^{-10}$ m/s^2 ('UST-K2' [1]). Equation (10) reproduces observed Abell cluster weak lensing shear profiles without requiring dark matter halo parameters.

6 Explicit Falsification Criteria

To maintain rigorous scientific standards, the cluster dynamics formulations in 'UST-K3' are subject to strict empirical falsification bounds.

Proposition 1 (Falsification Criteria for UST-K3)**.** *The substrate cluster dynamics model shall be considered falsified if any of the following observational conditions are established:*

1. ***Breakdown of the Relaxation Lag Scaling ($\Delta x \propto v \cdot \tau$):*** *If future high-velocity cluster collision surveys reveal spatial lensing offsets that systematically violate $\Delta x = R_{core}$ by more than 20%.*

2. ***Static Offsets in Relaxed Clusters:*** *If fully relaxed, non-colliding Abell clusters exhibit persistent spatial separations between baryonic mass centers and weak lensing peaks ($\Delta x_{offset} > 10$ kpc).*

3. ***Direct Detection of Particle Dark Matter:*** *If underground direct-detection experiments (e.g., LZ, SuperCDMS) or high-energy colliders discover WIMPs or sterile neutrinos possessing the cross-sections necessary to form cluster dark matter halos.*

7 Strategic Roadmap for the Cosmology Branch

With ‘UST-K1‘ [2] establishing cosmic redshift kinematics, ‘UST-K2‘ [1] establishing galactic rotation dynamics, and ‘UST-K3‘ establishing cluster dynamics, subsequent volumes address larger-scale cosmic phenomena:

- **UST-K4 (CMB Standing Wave Spectrum):** Formulates the CMB as thermal background equilibrium pressure ($P_0 \approx 1.516 \times 10^5$ N/m^2) and derives acoustic peak multipoles ($\ell_1 \approx 220, \ell_2 \approx 540, \ell_3 \approx 800$).
- **UST-K5 (Structure Growth & Crisis Resolutions):** Resolves the 5σ H_0 tension, JWST mature galaxies at $z > 10$, and the S_8 structure growth anomaly.

8 Conclusion

The Unified Substrate Theory demonstrates that gravitational lensing is the optical phase refraction of transverse shear waves passing through hydrostatic pressure shadow deficits ($n_{\text{refr}} = 1 + GM/c_s^2 r$). By deriving the Einstein deflection angle $\theta_{\text{deflect}} = 4GM/c_s^2 b$ and proving that the 200 kpc separation in the Bullet Cluster is an acoustic pressure relaxation lag ($\tau \approx 51.5$ Myr), UST-K3 provides a simple, zero-parameter ($k = 0$) continuum mechanism for cluster dynamics.

References

[1] R. Beem, *UST-K2: Substrate Cosmology II: Galactic Dynamics, First-Principles MOND Scale Derivation, and SPARC BTFR Benchmarks*, UST Cosmology Branch (2026).

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